

MACS+ - Membrane Action Of Composite Structures In Case Of Fire REPORT



1. Project properties

Project name:	Panattoni Crawley	Prepared by:	KevinKurniawan
Client name:	Lewis King	Company name:	Fire Dynamics
Client company:	Winvic Construction Limited	Date:	6 April 2025, 12:37
Job number:	0497	Checked by:	
Comments:	TMA Probabilistic Analysis for Plant Deck of Office A		

2. General arrangement

• Spans

Span 1: 7.75 m

Span 2: 8.64 m

• Unprotected beams

Number of internal unprotected beams: 2

3. Deck details

• Deck properties

Deck: ComFlor 60 Type: Trapezoidal
 Depth: 60 mm Top flange: 130.7 mm
 Pitch: 300 mm Bottom flange: 120 mm
 Stiffener height: 15 mm

4. Slab details

• Concrete

Concrete type: Normal Slab depth: 150 mm
 Cylinder compressive strength of concrete (f_{ck}): 30 N/mm²

• Mesh

Mesh type: A193
 Longitudinal mesh area: 193 mm²/m Bar size: 7 mm
 Transverse mesh area: 193 mm²/m Bar size: 7 mm
 Average mesh axis distance: 30 mm Mesh yield stress: 500 N/mm²

5. Beams details

• Unprotected beams

Beam type: Solid beam
 Section family: UK sections Steel grade: S355
 Degree of shear connection: 80 %
 Section size: UB 406x140x39
 Details: $h = 398 \text{ mm}$, $b = 141.8 \text{ mm}$, $t_w = 6.4 \text{ mm}$, $t_f = 8.6 \text{ mm}$

• Side A perimeter beam

Beam type: Solid beam
 Section family: UK sections Steel grade: S355
 Section size: UB 406x140x39
 Details: $h = 398 \text{ mm}$, $b = 141.8 \text{ mm}$, $t_w = 6.4 \text{ mm}$, $t_f = 8.6 \text{ mm}$
 Beam location: Internal beam Construction type: Composite
 Degree of shear connection: 80 %

• Side B perimeter beam

Beam type: Solid beam
 Section family: UK sections Steel grade: S355
 Section size: UB 457x152x52

Details:	h = 449.8 mm, b = 152.4 mm, $t_w = 7.6$ mm, $t_f = 10.9$ mm		
Beam location:	Edge beam	Construction type:	Composite
		Degree of shear connection:	80 %
• Side C perimeter beam			
Beam type:	Solid beam		
Section family:	UK sections	Steel grade:	S355
Section size:	UB 406x140x39		
Details:	h = 398 mm, b = 141.8 mm, $t_w = 6.4$ mm, $t_f = 8.6$ mm		
Beam location:	Internal beam	Construction type:	Composite
		Degree of shear connection:	80 %
• Side D perimeter beam			
Beam type:	Solid beam		
Section family:	UK sections	Steel grade:	S355
Section size:	UB 457x152x52		
Details:	h = 449.8 mm, b = 152.4 mm, $t_w = 7.6$ mm, $t_f = 10.9$ mm		
Beam location:	Edge beam	Construction type:	Composite
		Degree of shear connection:	80 %

6. Loading details

- Normal (Cold)
 - Leading variable action: 5 kN/m²
 - Accompanying variable action: 0 kN/m²
 - Dead load including beam, excluding slab: 0 kN/m²
 - Slab weight including mesh: 2.85 kN/m²
- Fire (Hot)
 - Combination factor for permanent action: 1.0
 - Combination factor for leading variable action: 0.5
 - Combination factor for other variable action: 0.3

7. Fire & Analysis

- Parametric temperature-time curve

Compartment length:	51.8 m	Window height:	4 m
Compartment width:	8 m	Window length:	51.8 m
Compartment height:	4 m	Glazing breakage:	64.641 %
Fire load:	415.84 MJ/m ²	Wall lining (B) factor:	1700 J/m ² s ^{1/2} K
Growth rate:	Medium	Combustion factor:	0.8

8. Summary output

- Default mesh direction

Maximum unity factor:	0.25	Floor slab adequate
Factored load in fire:	6.21 kN/m ²	
Fire curve:	Parametric temperature-time curve	

9. Default mesh direction

- Tabular results

Longitudinal mesh area:	193 mm ² /m	Bar size:	7 mm
Transverse mesh area:	193 mm ² /m	Bar size:	7 mm
Factored load in fire:	6.21 kN/m ²		

Time	Beam	Mesh	Slab top	Slab bottom	Beam capacity	Maximum allowable deflection	Slab yield	Enhancement	Slab capacity	Total capacity	Unity factor
mins	°C	°C	°C	°C	kN/m ²	mm	kN/m ²		kN/m ²	kN/m ²	
0	20	20	20	20	26.54	183	0.98	2.51	2.45	28.98	0.21
4	49	20	20	56	26.54	194	0.98	2.60	2.54	29.08	0.21

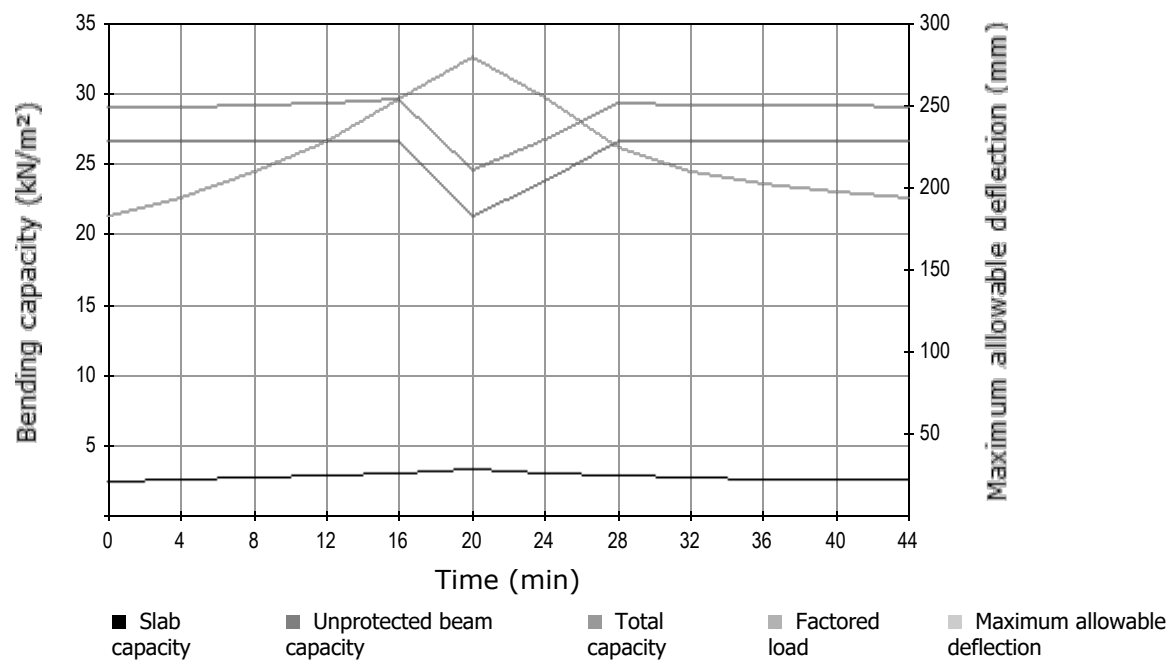
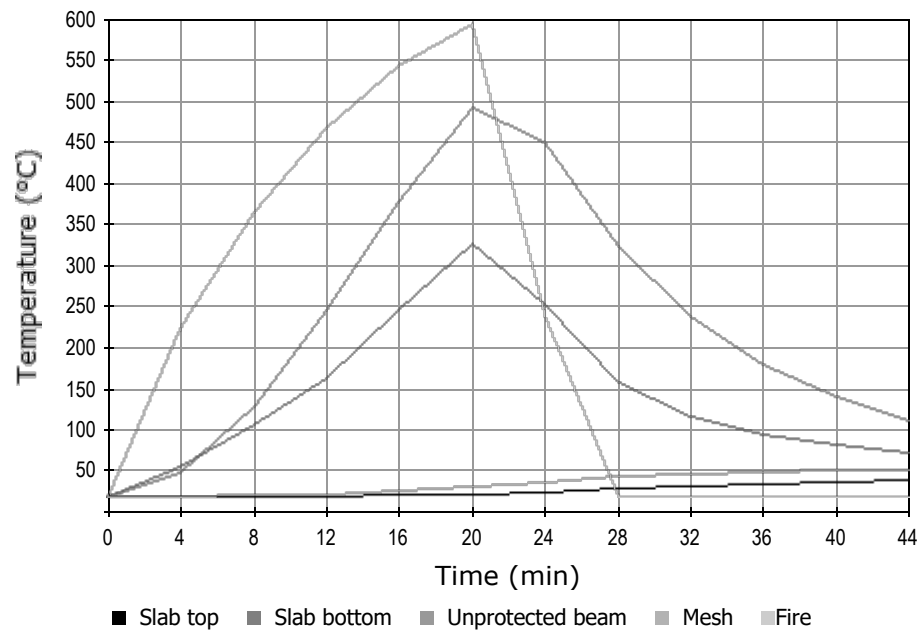
8	129	21	20	108	26.54	210	0.98	2.74	2.68	29.21	0.21
12	246	23	20	165	26.54	228	0.98	2.89	2.83	29.36	0.21
16	378	26	21	246	26.54	253	0.98	3.11	3.04	29.57	0.21
20	494	31	23	328	21.30	279	0.98	3.32	3.25	24.55	0.25
24	450	37	25	254	23.75	255	0.98	3.12	3.05	26.80	0.23
28	325	43	28	159	26.54	224	0.98	2.86	2.79	29.33	0.21
32	238	47	31	117	26.54	210	0.98	2.74	2.67	29.21	0.21
36	181	50	34	95	26.54	202	0.98	2.67	2.61	29.15	0.21
40	140	51	36	82	26.54	197	0.98	2.63	2.57	29.11	0.21
44	111	51	38	73	26.54	194	0.98	2.60	2.54	29.08	0.21

Maximum unity factor: 0.25 **Floor slab adequate**

• Perimeter beam check

Side A	Beam type:	Solid beam	Composite Internal beam
	Section size:	UB 406x140x39	
	Required moment resistance in fire situation:	0 kNm	
	Line load in fire situation:	0 kN/m	
	Shear connection:	80 %	
	Degree of utilization:	0.5	
	Critical temperature:	609 °C	
Side B	Beam type:	Solid beam	Composite Edge beam
	Section size:	UB 457x152x52	
	Required moment resistance in fire situation:	0 kNm	
	Line load in fire situation:	0 kN/m	
	Shear connection:	80 %	
	Degree of utilization:	0.5	
	Critical temperature:	611 °C	
Side C	Beam type:	Solid beam	Composite Internal beam
	Section size:	UB 406x140x39	
	Required moment resistance in fire situation:	0 kNm	
	Line load in fire situation:	0 kN/m	
	Shear connection:	80 %	
	Degree of utilization:	0.5	
	Critical temperature:	609 °C	
Side D	Beam type:	Solid beam	Composite Edge beam
	Section size:	UB 457x152x52	
	Required moment resistance in fire situation:	0 kNm	
	Line load in fire situation:	0 kN/m	
	Shear connection:	80 %	
	Degree of utilization:	0.5	
	Critical temperature:	611 °C	

10. Graphical output



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